

Integrals – 4

Series and integrals

1 сентября 2026 года

Theoretical minimum.

Theorem 1 (termwise integration). If $\sum f_n$ converges uniformly on $[a, b]$, or is dominated by an integrable function, then $\int_a^b \sum f_n = \sum \int_a^b f_n$. For series of constant sign the monotone convergence theorem suffices; for alternating series, the Leibniz test together with the estimate of the remainder.

Theorem 2 (the sophomore's dream, Bernoulli 1697). $\int_0^1 x^{-x} dx = \sum_{n=1}^{\infty} n^{-n}$ and $\int_0^1 x^x dx = \sum_{n=1}^{\infty} (-1)^{n+1} n^{-n}$.

Fact 1. The dictionary: $\int_0^1 \frac{\ln x}{1-x} dx = -\frac{\pi^2}{6}$, $\int_0^1 \frac{\ln x}{1+x} dx = -\frac{\pi^2}{12}$, $\int_0^{\infty} \frac{x^{s-1}}{e^x-1} dx = \Gamma(s)\zeta(s)$; the last formula is the bridge between integrals and the zeta function.

1. Prove both formulas of Theorem 2, justifying the interchange of sum and integral, and compute $\int_0^1 x^{-x} dx$ numerically to three decimal places.
2. Prove that $\int_0^{\infty} \frac{x^{s-1}}{e^x-1} dx = \Gamma(s)\zeta(s)$ for $\operatorname{Re} s > 1$, and deduce $\int_0^{\infty} \frac{x}{e^x-1} dx = \frac{\pi^2}{6}$ and $\int_0^{\infty} \frac{x^3}{e^x-1} dx = \frac{\pi^4}{15}$.
3. (Integration Bee) A series under the integral sign:
 - a) $\int_0^{1/2} \sum_{n \geq 0} \binom{n+3}{n} x^n dx$ (2022);
 - b) $\int_0^1 \sum_{k \geq 1} (-1)^k x^{2k} dx$ (2025);
 - c) $\int \sum_{k \geq 0} \sin\left(\frac{k\pi}{2}\right) x^k dx$ (2024);
 - d) $\int_0^{1/2} \sum_{n \geq 2} x^n dx$ (2026).

In each case first find the sum of the series in closed form.

4. (Integration Bee 2023) Compute $\int_0^1 \prod_{k=0}^{\infty} \frac{1}{1+x^{2^k}} dx$. (*Hint: multiply out the first few factors and recall binary expansions.*)
5. Compute $\int_0^1 \ln x \ln(1-x) dx$ and $\int_0^1 \frac{\ln^2 x}{1-x} dx$ by expanding one of the factors into a series; the answers should involve $\zeta(2)$ and $\zeta(3)$.